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RESEARCH ARTICLE

Toxicity of urban highway runoff in Shanghai to Zebrafish (*Danio rerio*) embryos and luminous bacteria (*Vibrio qinghaiensis.Q67*)

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Abstract Pollution from urban highway runoff has been identified as one of the major causes of the deterioration of receiving water quality. The purpose of this study is to assess the toxicity of urban storm water samples in Shanghai using the zebrafish (*Danio rerio*) embryo test and the bacterial luminescence (*Vibrio qinghaiensis*) assay. The toxicity of highway runoff from seventeen storm events was investigated in both grab and composite samples. Zebrafish embryos were exposed to the runoff samples and development parameters including lethality, spontaneous movements in 20 s, heart beat rate, hatching rate, and abnormality of zebrafish embryos were observed after 24, 48, 72, and 96 h of exposure. Inhibition rates of luminescence intensity were also recorded. The results showed that in the zebrafish embryo toxicity tests, both grab and composite samples increased the lethality, reduced the percentage with spontaneous movements and heart beats, inhibited the hatching of embryos, and induced morphological abnormalities. In the *Vibrio qinghaiensis* toxicity test, all the grab samples inhibited the luminescence, while some of the composite samples promoted it, which indicated that different types of toxicants might have been affecting the species. The multivariate statistics analysis indicated that heavy metal (zinc, manganese, and copper) and PAHs might mainly contribute to the toxicity of runoff samples.

Keywords Urban highway runoff · Zebrafish embryos · Luminous bacteria · Toxicity · Heavy metal

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Introduction

Non-point pollution from urban highway runoff has been identified as one of the major causes of the deterioration of the receiving water quality (Lee and Bang 2000). The component of pollutants in the urban highway runoff is very complex, which may include as many as 600 different xenobiotic compounds (Eriksson et al. 2007). These compounds threaten aquatic organisms and contribute to the alternation of freshwater ecosystems (Waara and Farm 2008; Clément et al. 2010). Numerous factors may affect the quality of urban runoff, such as traffic volume, antecedent dry period, rainfall intensity, and duration (Gunawardena et al. 2013; Liu et al. 2012; Memon et al. 2013). The constituents of urban runoff include primarily high concentrations of nutrients, heavy metals and polycyclic aromatic hydrocarbons (PAHs), as well as herbicides (McKenzie and Young 2013; Stein et al. 2012).

Studies have shown that toxicity of runoff samples possibly associates with heavy metals and PAHs due to tires and exhaust gas of vehicles on the urban highway (Karlsson et al. 2010; Kayhanian et al. 2008; Brown and Peake 2006). In most studies, urban highway runoff samples were tested for the ecotoxicity with different organisms, for instance, *Daphnia magna*, purple sea urchin, luminescent bacteria, and fish embryos (Waara and Farm 2008; McQueen et al. 2010; Dorchin and Shanas 2010). Fish embryos are increasingly being used as a model in ecotoxicology in the recent years (Nagel 2002; Scholz et al. 2008;Knobel et al. 2012; Heger et al. 2012). The zebrafish embryos toxicity (FET) test, as an OECD guideline, has been

an alternative for the adult fish toxicity test (Strahle et al. 2012). Zebrafish embryos have been widely used to test the toxicities of environmental relevant contaminants (Ho et al. 2013; van Aerle et al. 2013; Fouqueray et al. 2013), polluted water (Chen et al. 2007; VanLandeghem et al. 2012), waste water (Sisman et al. 2008; Karadima et al. 2009; Hafeli et al. 2011), suspended particulate (Keiter et al. 2006; Rocha et al. 2011), whole sediments (Kosmehl et al. 2008, 2012; Wu et al. 2010), and sediment eluates or extracts (Kosmehl et al. 2008; Wu et al. 2010; Higley et al. 2012). Studies with zebrafish embryo indicated that this biomarker is very easy to use, practical, rapid, and sensitive for assessing aquatic quality (Hallare et al. 2005). The Microtox assay system is another rapid and cost-effective bioassay, which is based on the measurement of reduction in light emission of the luminescent bacteria after exposure to the toxicant. The freshwater luminescent bacteria (*Vibrio qinghaiensis sp. Nov.* 67 (hereafter referred to as *Q67*)) exhibit a wide pH tolerance and can be luminous in freshwater (Ma et al. 1999). It has been widely used as reproducible and sensitive screening method to determine overall toxicity of various types of samples (Ma and Wang 2013; Ma et al. 2012).

With the rapid process of urbanization in recent years, the urban highway system has been continuously expanding in Shanghai, resulting in increasingly serious runoff pollution (Ballo et al. 2009). The highway in Shanghai has reached 80 km (Li and Zou 2004), with traffic flow of about 2 million pcu/day. The main objective of this study is to characterize the potential toxicity of highway runoff during 17 rainfall events in Shanghai by using the approaches of zebrafish embryo toxicity tests and luminescent bacteria *Q67* acute toxicity tests. The study will not only provide a basis for runoff pollution control and detoxification in particular in Shanghai, but also contribute to the database of ecotoxicity study of urban highway runoff.

Methods

Sites and sampling

The sample site is located at a highway in a highly urbanized area of Shanghai (Fig. 1). Highway runoff samples were collected from seventeen rainfall events during April–August 2011, with the traffic volume between 4,154 and 9,820 vehicles/h.

Both grab and composite samples were collected to determine the toxicity of highway runoff. Garb samples were used to investigate the toxicity characteristic of highway runoff with the rainfall duration, and composite samples were used to evaluate the potential toxic effect of the highway runoff. Grab samples were collected in three storm events on April 15, May 22, and June 4, respectively, in 2011, with runoff duration of 60, 30, and 30 min, respectively. Grab samples were collected every 15 min. Fourteen storm events were monitored (with composite

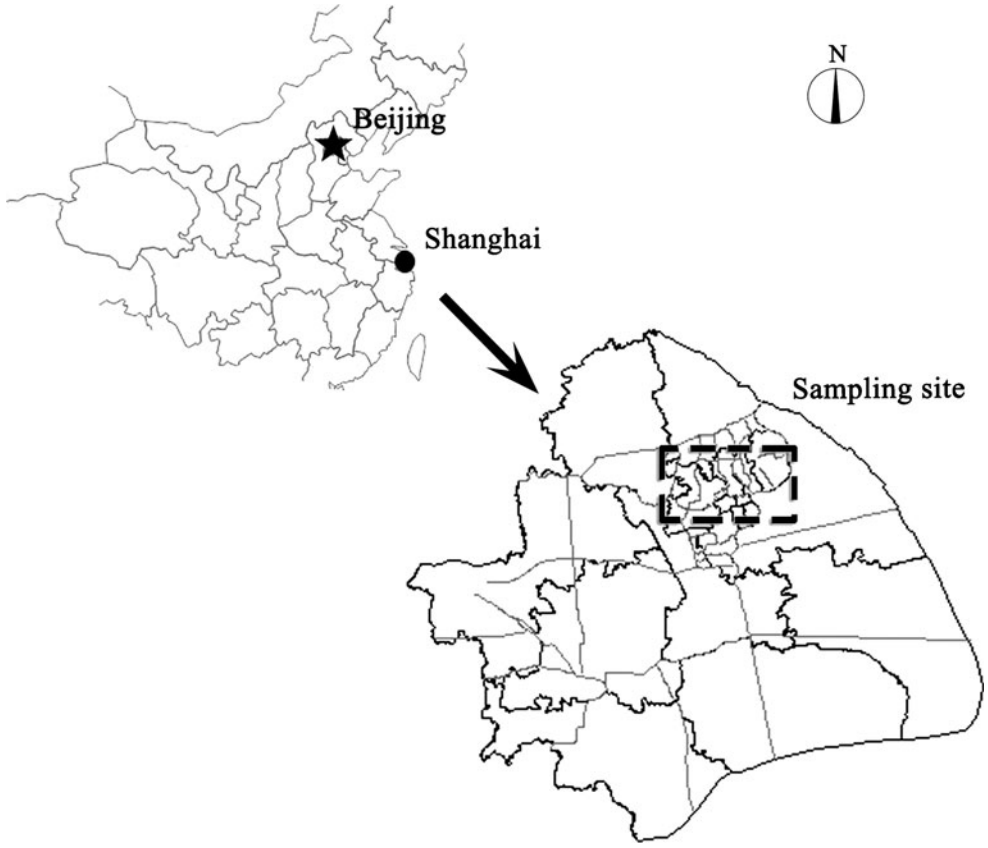

Fig. 1 Location of sampling site in Shanghai

samples being taken) on May 23, June 6, 10, 14, 15, 17, 18, and 21, July 4, 14, and 31, and August 3, 4, and 14 in 2011, respectively.

The runoff samples for heavy metal analysis were collected in polyethylene containers (1 l) placed at the bottom of storm-drains on both undersides of the highway. Samples for PAHs analysis and toxicity tests were collected in glass bottles (1 l) with polypropylene caps. All samples were filtered with a 0.45-μm mesh glass microfiber membrane and stored immediately in the dark at 4 °C until analyzed. Chemical analysis and toxicity tests were initiated within 36 h after sample collection.

Chemical analysis

The runoff samples were acidified prior to measurement of heavy metals using HCl/HNO_3. The concentrations were determined by ICP-AES (Optima 2100 DV, Perkin Elmer, USA) for the element composition.

PAHs were extracted using a solid phase extraction (SPE) system from Supelco (Sigma-Aldrich Co, USA). The SPE column (Supeclean Envi-C18) were firstly activated with 10 ml of dichloromethane and then washed with 6 ml of methanol and 6 ml of ultra-pure water. Water samples (500 ml) passed through the column at a flow rate of 5 ml/min under vacuum pump. After extraction, the columns were eluted with 6 ml dichloromethane and then combined with dichloromethane rinse from the extraction column. The elution of water sample was reduced to 0.5~1.0 ml under a gentle flow of high-purity nitrogen, subject to a solvent exchange to methanol, and concentrated to 0.5 ml by the rotary evaporator prior to chemical analysis.

The determination of PAHs was performed with GC-MS. The main parameters and the determination conditions were as follows: capillary column of 30 m×0.325 mm×0.25 μm for GC: carrier gas of the helium (purity 99.99 %) with flow rate of 0.8 ml/min; 2.0 μl volume of inject volume in a splitless mode and injector temperature of 290 °C. The column temperature was initially controlled at 70 °C for 4 min, ramp at 8 °C/min to a final temperature of 300 °C (hold for 5 min). Full scanning of the samples was carried out and the electron impact energy was set to 70 eV. The temperature of ion source and the connection device were 200 °C and 290 °C, respectively. Quantitative and qualitative determination of the PAHs were made by comparison of relative retention time and peak area of mass spectrogram between reference materials and real water samples.

Acute toxicity test with zebrafish embryos

Zebrafish is a small benthopelagic cyprinid fish with a mean adult length between 3 and 5 cm. They were raised in 50 l tanks in the laboratory with males and females kept at a ratio of 1:2 under the controlled conditions of temperature (26.0±1.0 °C), hardness (250 mg/l $CaCO_3$), pH (7.8±0.2), and dissolved oxygen (≥80 % saturation). The fish were fed with commercially frozen bloodworms sterilized by ultraviolet lamps twice per day. The photoperiod was adjusted to 14 h light and 10 h darkness.

After culturing for 3 months, mature zebrafish can be used in the tests with the mortality under 10 %. Mature zebrafish were put together in spawning box, with the ratio of male and female was 1:2. Spawning and fertilization took place within 30 min after light had been turned on in the morning. About 30 min after spawning, the eggs were collected and rinsed with the reconstituted water (ISO 7346/3: 294 mg/l $CaCl_2 \cdot 2H_2O$, 123 mg/l $MgSO_4 \cdot 7H_2O$, 123 mg/l $NaHCO_3$, 5.5 mg/l KCL) that had been aerated to nearly 100 % oxygen saturation prior to bioassay. Eggs were then screened to ensure that they were fertilized and developing normally by the inverted microscope (Nikon TE2000-U). For control experiments, a fertilization rate should be higher than 90 %.

The acute toxicity test method used in the present study was a modification of the FET test (OECD 2013). Fertilized eggs were exposed in 24-well cell culture plates with a Pasteur pipette. 2 fertilized eggs were placed in one well with 2 ml of

Table 1 Chemical analyses of the runoff samples from seventeen rainfall events

		CMC[a]	CCC[b]	Garb samples		Composite samples	
				Mean	SD	Mean	SD
Heavy metals (μg/l)	As	340	150	12.8	10.6	21.6	13.7
	Se	13~186	5	18.3	22.8	7.4	8.1
	Zn	120	120	304.6	189.8	126.8	118.1
	Pb	65	2.5	5.1	0.8	9.5	11.9
	Cd	2.0	0.25	—	—	—	—
	Ni	470	52	12.7	0	8.5	6.4
	Fe	—	1,000	135.3	89.0	84.7	58.6
	Mn	2,300	80	160.3	120.0	50.6	89.8
	Cr	—	—	—	—	6.2	2.1
	Cu	13	9	102.6	71.3	28.7	32.8
	Al	750	87	168.2	33.8	240.6	90.4
	Fluorene	70	3.9	27.4	9.6	20.7	5.4
LMW PAHs (μg/l)	Phenanthrene	30	6.3	41.6	12.5	21.7	6.6
	Anthracene	13	0.73	13.8	10.8	14.1	2.9
	Pyrene	—	0.025	2.1	1.1	—	—
	∑16PAHs[c]	—	—	84.2	26.2	40.7	5.7

[a] The criterion maximum concentration from National Recommended Water Quality Criteria (NRWQC) for evaluation of contamination in the urban stormwater runoff (US EPA 2009). The CMC representing acute exposures

[b] The criterion continuous concentration from National Recommended Water Quality Criteria (NRWQC) for evaluation of contamination in the urban stormwater runoff (US EPA 2009). The CCC representing chronic exposures

[c] Sum of concentration of naphthalene, acenaphthylene, acenaphthene, fluorene, phenanthrene, anthracene, fluoranthene, pyrene, benzo(a)anthracene, chrysene, benzo(b)fluoranthene, benzo(k)fluoranthene, benzo(a)pyrene, indeno(1,2,3-cd)pyrene, dibenzo(a,h)anthracene and benzo(g,hi)perylene

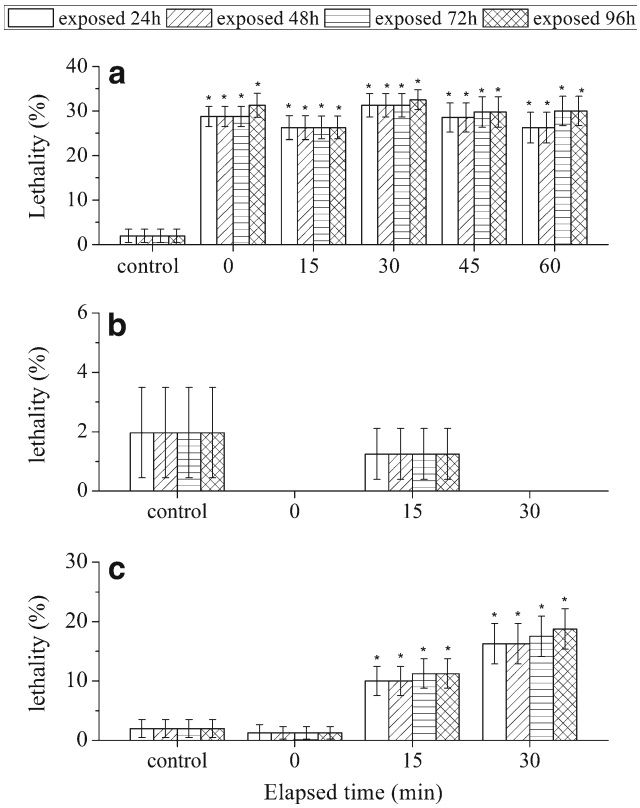

Fig. 2 Zebrafish embryos lethality at 24-h exposure response to elapsed time (grab samples, three storm events in April 15–June 4, 2011. **a** April 15. **b** May 22. **c** June 4. *Asterisks* indicate significant difference from the control ($P<0.05$))

test solutions amounting to a total of 40 eggs per test. The remaining four wells of each plate were used as control with the reconstituted water. The 24-well-plates were covered with foil and incubated at 26.0±1.0 °C and 14 h light/10 h dark photoperiod. The development of embryos from blastula to early larval stage was monitored at specific time points (24, 48, 72, and 96 h). Endpoints used for assessing the embryotoxicity included lethality, spontaneous movement, heartbeat, hatching success, and abnormality. These lethal and sub-lethal endpoints were observed and recorded for both reference and treated group using an inverse microscope. Test solutions were replaced daily with freshly saturated solutions and the embryos were not fed during the test. The tests had three replicates including the control group exposed to zebrafish embryos.

Q67 acute toxicity test

For the acute toxicity test, luminescent bacterium *Q67* (purchased from Beijing Hammatsu Photon Techniques Inc.) freeze-dried as pellets in glass bottles were removed from −20 °C storage before the test, with recovery liquid (0.8 % NaCl) added for rehydration at 20 °C for 15 min. The standard methods for culture medium preparation and *Q67* incubation were adopted by referring to the previous articles (Shao et al. 2012). The luminescence inhibition assay was performed in test tubes using a luminometer (Model RS9901, Shanghai Rongsheng Biological Electronics Co.). For each test, four test tubes were prepared, three for parallel samples and one for blank control (recovery liquid). *Q67* were directly exposed to the runoff samples. Samples or control liquid (2 ml) were added into each tube and the bacterial suspension (50 μl) was added at a 10 s interval. After 15 min exposure of the bacteria to the sample at 20 °C, the relative light unit of luminescent bacterium *Q67* was measured. The toxicity was evaluated by the relative inhibition of light emission. Each test was repeated three times, and the average inhibition ratio was taken as the final result.

Data analysis

Significant ($P<0.05$) data were obtained by using *t* test, expressed as mean ± SD. The effect of different factors in

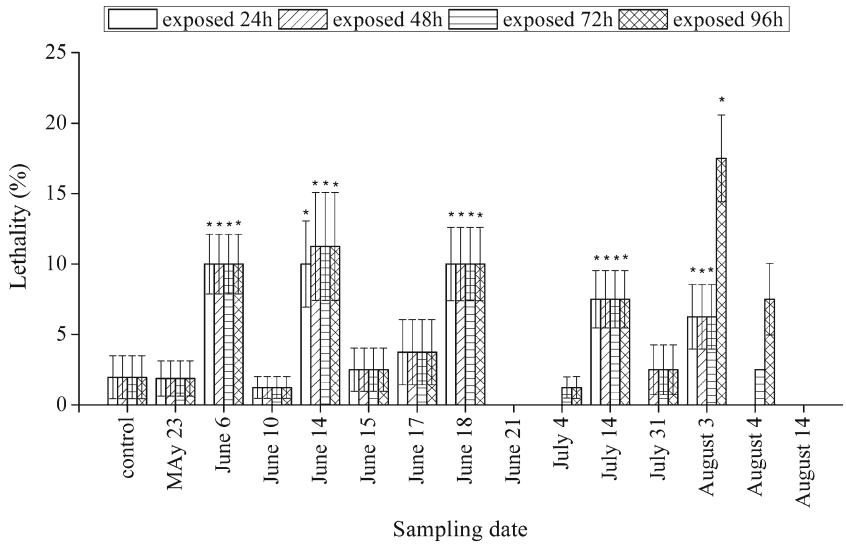

Fig. 3 Lethality of zebrafish embryos exposed to composite samples of urban stormwater runoff for 96 h (May 23–August 14, 2011 storm events. *Asterisks* indicate significant difference from the control ($P<0.05$))

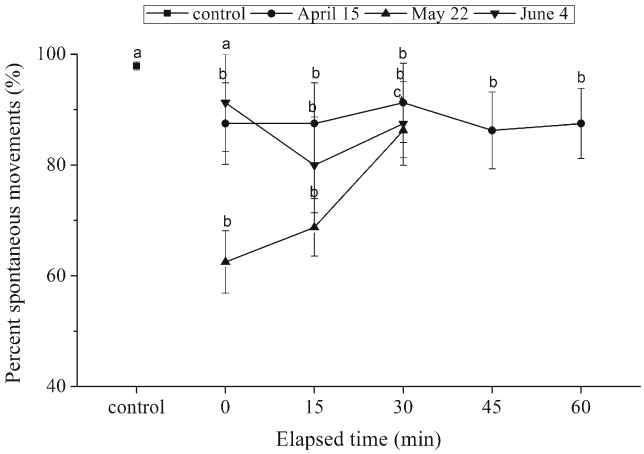

Fig. 4 Zebrafish embryos spontaneous movements in 20 s at 24-h exposure response to elapsed time (grab samples, April 15–June 4, 2011 storm event. Spontaneous movements with a *different letter* are significantly different in the same date ($P<0.05$))

stormwater runoff on embryos was tested using one-way analysis of variance (ANOVA) followed by Dunnett's multiple comparison tests (when data were normally distributed according to least-significant difference tests). All the statistical analysis was carried out using SPSS 17.0 Statistical Software.

Results

Chemical parameters and sample characteristics

Highway runoff samples were collected from April 15 to August 14, 2011 during 17 rainfall events, including eleven grab samples and 14 composite samples. The investigation focused on analytical measurements of major trace metals and PAHs in the urban highway runoff. The concentrations of these pollutants in all the samples are shown in Table 1.

High concentrations of heavy metals Zn, Fe, Mn, Cu, and Al were found in all samples. Concentrations of these chemicals were compared with the criterion maximum concentration (CMC) and criterion continuous concentration (CCC) from National Recommended Water Quality Criteria (NRWQC) for evaluation of contamination in the urban stormwater runoff (US EPA 2009). The CMC is known as the "acute" aquatic life ambient water quality criterion. The pollutants, concentrations of which have exceeded the value of CMC, indicate that adverse effects on living creatures would occur frequently in a short time. The CCC is equated to chronic toxicity. The pollutants, concentrations of which have exceeded the value of CCC, indicate that adverse effects on living creatures might increase in incidence from rare to occasional during a long period of time. The heavy metals Cd was not detectable, whereas the concentrations of Se, Pb, Mn, and Al exceeded CCC in most runoff samples. It is noticed that the concentrations of Zn and Cu exceeded CMC in all the samples.

In this investigation, the concentrations of low-molecular weight (LMW) PAHs in all the runoff samples were lower than CCC. Therefore, heavy metals were regarded as the main pollutants in the urban highway runoff.

Lethality to zebrafish embryos

The lethality of zebrafish embryos after the grab samples exposure is shown in Fig. 2. The embryos lethality was less than 5 % in the control group. Compared with the control group, the grab samples of the storm discharge were toxic to the embryos ($P<0.05$, Fig. 2a, c). More significant egg coagulation phenomenon was observed after the 24-h exposure in three storm events, with

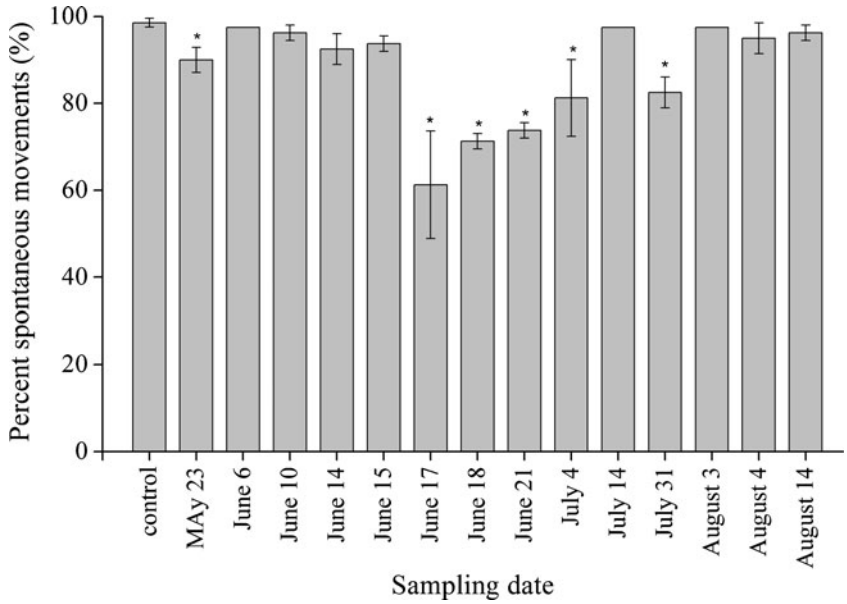

Fig. 5 Percentage of zebrafish embryos spontaneous movements in 20 s at 24-h exposed to composite samples of urban stormwater runoff (May23–August14, 2011 storm events. *Asterisks* indicate significant difference from the control ($P<0.05$))

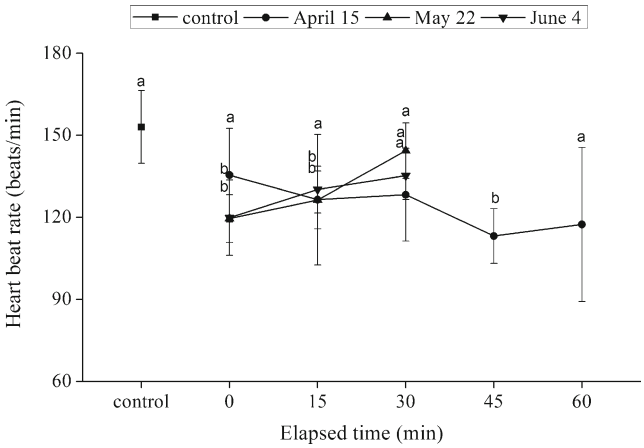

Fig. 6 Zebrafish embryos (48 hpf) response to elapsed time (grab samples, April 15–June 4, 2011 storm event. Heart beat rates with a *different letter* are significantly different in the same date ($P<0.05$))

the lethality of 0~31.25 (±2.63) % (Fig. 2a, b, c). However, there was no significant relationship between the rainfall duration and lethality. The lethality of the zebrafish embryos at 96-h exposure in the composite samples varied from 0~17.5 (±3.07) %, which is significantly higher than the control group ($P<0.05$; Fig. 3). Moreover, during 96-h exposure, the lethality did not obviously rise with the exposed time.

Sublethality of zebrafish embryos

Spontaneous movements

The spontaneous movements were not less than once per 20 s with 24-h exposure for the normal developed zebrafish embryos. It can be seen from Figs. 4 and 5 that compared with the control group, for most of the grab samples and composite samples the percentage with spontaneous movements is significantly smaller ($P<0.05$). For the grab samples, in particular, the percentage was significant lower than the control at the beginning of rainfall for the three storm events, followed by rising with elapsed time from 15 to 30 min, but again decreasing in the following 30 to 60 min elapsed time (Fig. 4). However, for the composite samples, after exposure to the stormwater runoff, the percent with spontaneous movements varied from 61.25 % to 97.5 % (Fig. 5).

Heart beat rate

The heart beat rates of the zebrafish embryos exposed to the grab samples and composite samples are shown in Figs. 6 and 7. After exposure to the grab samples of three storm events, the heart beat rates (48 hpf) of zebrafish embryos were lower than the control group (Fig. 6), especially for those exposed to the runoff collected at the initial phase, the heart beat frequency reduced significantly ($P<0.05$). After exposure to the composite samples, except exposure in the storm events in June 17, the heart beat frequency had no significantly difference with the control group $P<0.05$ (Fig. 7).

Hatching rate

The hatching rate of the zebrafish embryos was 98.62 (±1.40) % after 72 h in the control group. After 72-h exposure to the grab samples, the embryos exhibited hatching delay (Fig. 8—1).

The hatching rate reduced after 72-h exposure in the three storm events (Fig. 8—1). After exposure to the runoff collected at the initial phase, the hatching rate of the zebrafish embryos significantly reduced ($P<0.05$), especially for those exposed

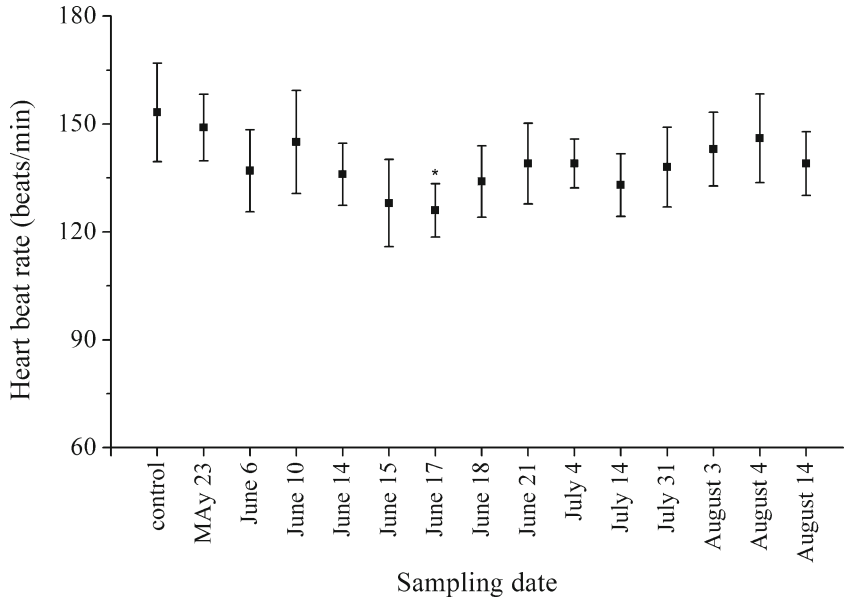

Fig. 7 The 48-h heart beat rate of zebrafish embryos exposed to composite samples of urban stormwater runoff (May 23–August14 2011 storm events. *Asterisks* indicate significant difference from the control ($P<0.05$))

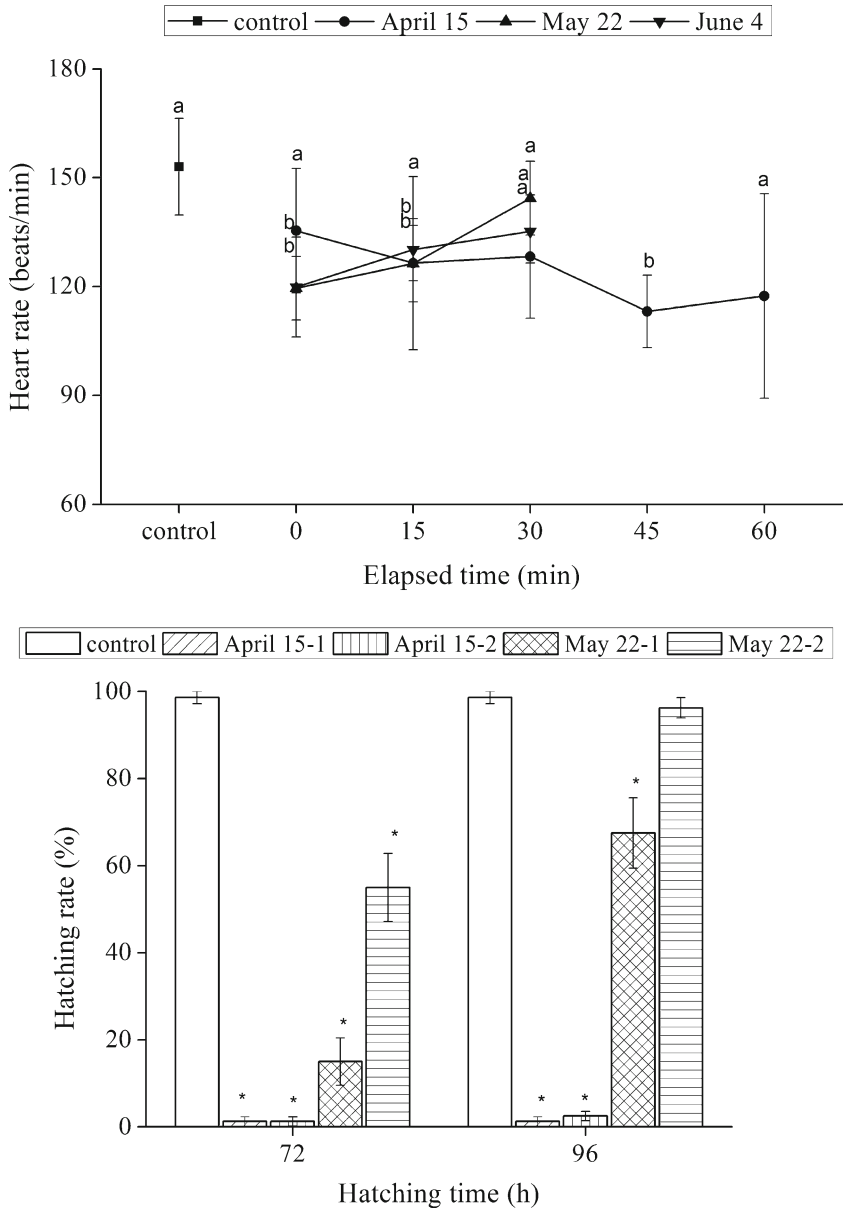

Fig. 8 *1* Zebrafish embryos hatching rate (72 hpf) with elapsed time (grab samples, April 15–June 4, 2011 storm events. Hatching rates with a *different letter* are significantly different in the same date ($P<0.05$)). *2* Zebrafish embryos hatching rate (72 hpf) with hatching time (April 15—*1* elapsed 0 min runoff samples on April 15, April 15—*2* elapsed 15 min runoff samples on April 15; May 22—*1* elapsed 0 min runoff samples on May 22, May 22—*2* elapsed 15 min runoff samples on May 22. *Asterisks* indicate significant difference from the control ($P<0.05$))

to the samples taken on April 15 and May 22, which was lower than 20 % after 72 h-exposure (Fig. 8—2), but the hatching rate increased with the elapse of time (Fig. 8—1). After 72-h exposure to composite samples (such as June 6, 14, 18, July 14, and August 3), the hatching rates reduced significantly compared with the control group ($P<0.05$; Fig. 9). However, no significant difference in the hatching rates was observed after 72-h exposure to other composite samples from urban highway runoff (Fig. 9).

Teratogenesis

After 96-h exposure to urban highway runoff, some development abnormalities of the zebrafish embryos were observed, including pericardial edema, blood clotting, and lordosis (Fig. 10). In the experimental group of grab samples, the percentage of abnormalities observed among zebrafish embryos decreased with the elapse of time in the first 30 min of storm events (Fig. 11). The abnormality rates in the composite samples varied from 1.75 % to 10 %, except for the composite samples taken on August 3 (with the abnormality rate up to 55 %; Fig. 12).

Inhibition rate of *Q67*

The acute toxicity effects of urban stormwater runoff on *Q67* were observed mainly from the inhibition of the luminous intensity, ranging from 0.96 (±0.63) %~60.99 (±6.06) % (Fig. 13). The inhibition rate was high at the beginning, but decreased significantly in 0 to 15 min ($P<0.05$), which indicated that the toxicity effect of runoff on *Q67* was more

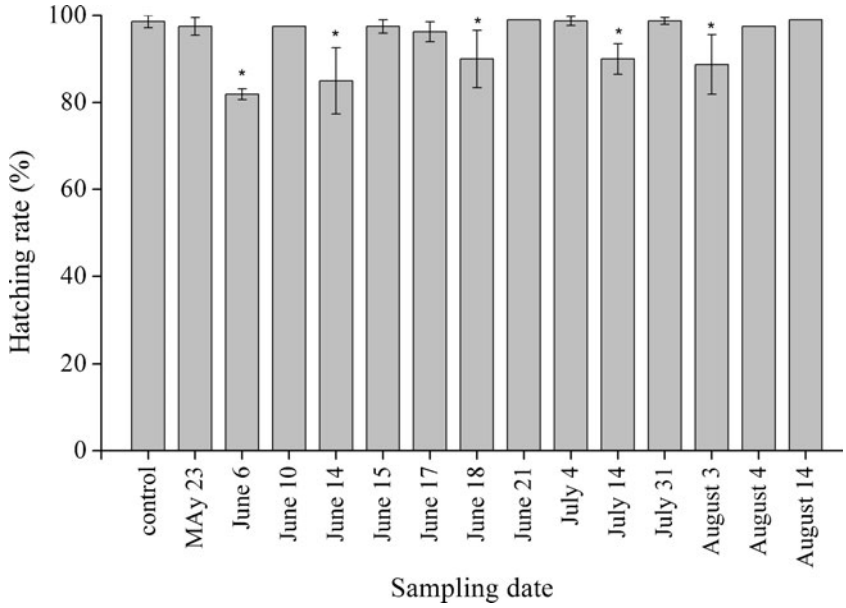

Fig. 9 Hatching rate of zebrafish embryos (72 hpf) exposed to composite samples of urban stormwater runoff (May 23–August 14, 2011 storm events. *Asterisks* indicate significant difference from the control ($P<0.05$))

obvious with the runoff collected at the initial rainfall phase. In addition, *Q67* luminous intensity increase (inhibition rates −1.66 (±1.24)~−25.09 (±2.71) %) was observed from some exposed composite samples (Fig. 14), the phenomenon of

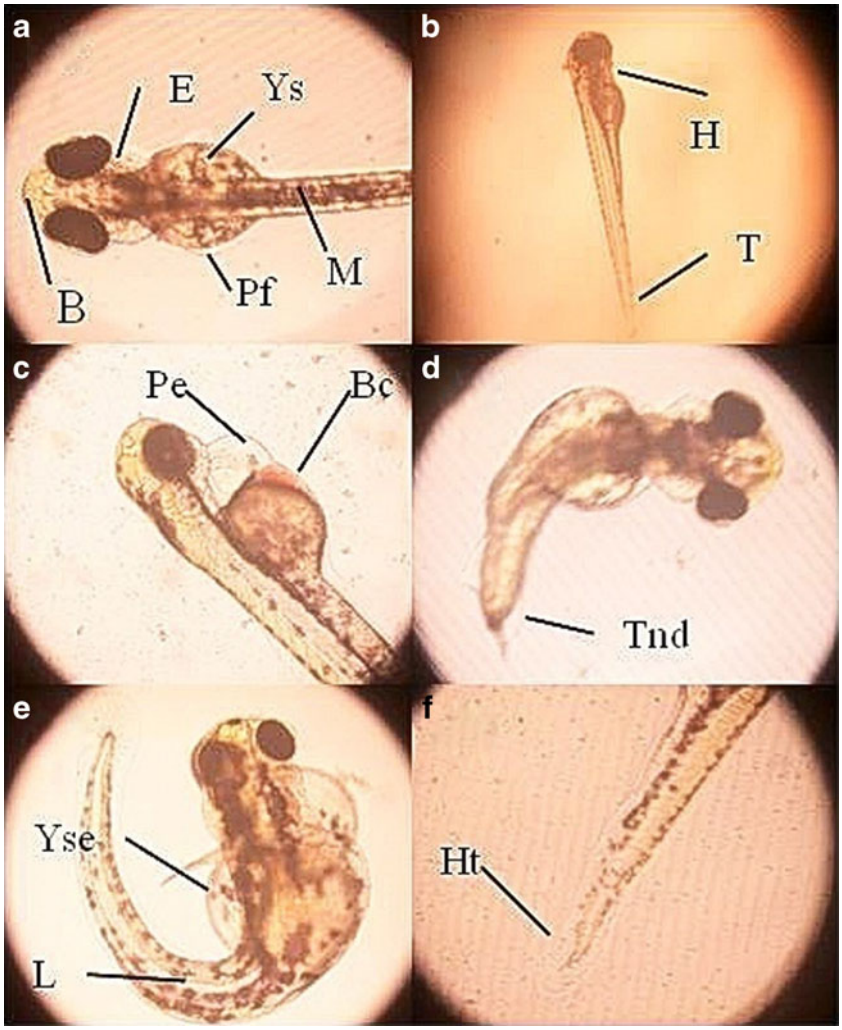

Fig. 10 Abnormalities of zebrafish embryos exposed to urban stormwater runoff. Note: **a**, **b** normal embryo after 96 h, **c** pericardial edema and blood clotting, **d** undetached tail, **e** yolk sac edema and lordosis, and **f** underdeveloped tail. *B* brain, *Bc* blood clotting, *E* eye, *H* heart, *Ht* helical tail, *L* lordosis, *M* melanophores, *Pe* pericardial edema, *Pf* pectoral fin, *T* tail, *Tnd* tail not detached, *Ys* yolk sac, *Yse* yolk sac edema

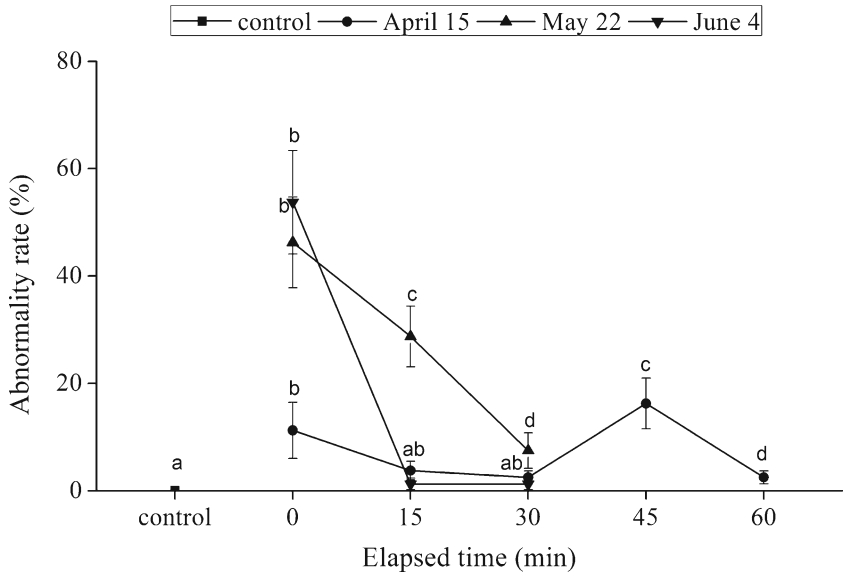

Fig. 11 Zebrafish embryos abnormality rate (96 hpf) with elapse of time (grab samples, April 15–June 4, 2011 storm event. Abnormality rates with a *different letter* are significantly different in the same date ($P<0.05$))

which was known as "hormesis-effect," which was influenced by the low concentration of toxins caused by bacterial metabolism. The acute toxicity of runoff to *Q67* showed that the runoff collected the initial phase had more significant toxic effect than that at later stage. However, similar hormesis regularities were not observed for *V. qinghaiensis Q67*.

Discussion

The CMC and CCC value have been widely used to assess the water quality. According to the chemical analysis of urban stormwater runoff, the concentrations of copper and zinc were found exceeding the CMC value. The concentrations of selenium, lead, manganese, and aluminum also exceeded the CCC value in some runoff samples. Therefore, heavy metals might be the key pollutants from urban highway runoff in this study.

The results of zebrafish embryos toxicity test suggested that both grab and composite samples increased the lethality, reduced heart beats and percentage with spontaneous movements, inhibited the hatching of embryos, and induced development abnormalities. Most zebrafish embryos exposed to the composite runoff samples could continue developing and hatch out from membrane, but the activity of the larvae was poor. It is found that the toxic effect of grab samples in the present study is much less than that of the surface water from the urban and rural river basins in Texas (VanLandeghem et al.

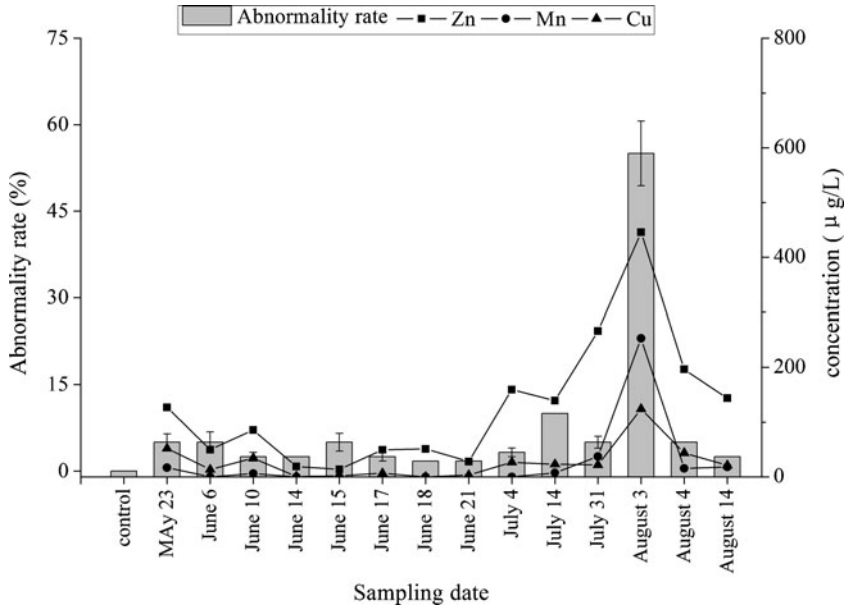

Fig. 12 Percentage of abnormalities observed among zebrafish embryos exposed to composite samples and the corresponding concentration of Zn, Mn, and Cu (May 23–August 14, 2011 storm events)

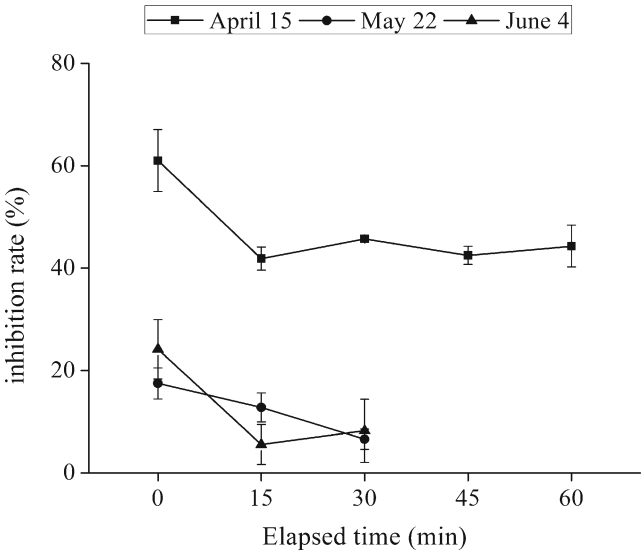

Fig. 13 Influence of runoff elapsed time on inhibition rate of *Q67* (April 15–June 4, 2011 storm events)

2012), but it is close to that of water samples organic extracts from Lake Taihu, China, in which organics were concentrated by 31.25 (Chen et al. 2007). It can be inferred that the urban highway runoff could cause to the deterioration of the receiving water quality. Data from statistical analysis showed that, the dominating factors affecting the lethality of zebrafish embryos were zinc and iron (Table 2). The key factors influencing sublethal effect, especially the hatching rate and abnormal development of zebrafish embryos, were zinc, manganese, copper, PAHs, etc. (Fig. 12). For example, the highest abnormality rate was observed in the runoff sample from the August 3 event with the highest measured concentrations of Zn, Mn, and Cu.

Heavy metals can pass through the chorion and cause direct and visible injury to the embryo, i.e. the eye and otolith agenesis, melanin cell growth retardation, atrial edema, heart cyst edema, spinal deformity, and tail deformity. Cu can not only lead to the deformity of embryo and larvae, but also induce spirality of the nervous system (Ozoh 1979). The relative coefficient between hatching rate and abnormality percentage and Cu concentration were −0.583 ($p<0.01$) and 0.881 ($p<0.01$), respectively (Table 2). However, there was no significant correlation between Cu concentration and lethality (24 hpf). A review of toxicity of copper to zebrafish embryos and larvae has revealed that hatching is more sensitive to copper than larval survival, and the highest concentration (128 μg/l) kills all embryos within 24 h, while survival is not affected in the lowest concentration (0.125 μg/l; Dave and Xiu 1991). In this study, the relatively high concentration of copper (from 0.67 to 229.33 μg/l) did not induce the high toxic effect. Therefore, the toxic effect of runoff samples on zebrafish embryos may result not from only one mechanism, but also from a combination of them through antagonism or synergism.

Of the heavy metals, zinc in minute quantities is considered to have various functions in biological development such as cell structure, enzyme activities, protein, and carbohydrate metabolism of fish (Williams and Holdway 2000). However, excessive zinc may induce toxic effects on the embryo and larval development. For example, poor survival, delayed or promoted time-to-hatch, low hatchability, reduced size of newly hatched larvae, and abnormality due to zinc exposure has been observed in many fish, including zebrafish (Dave et al. 1987; Jezierska et al. 2009). In the present study, there is significant correlation between lethality, hatching rate and abnormality percentage and Zn concentration (Table 2). Meanwhile, the significantly relative coefficient between

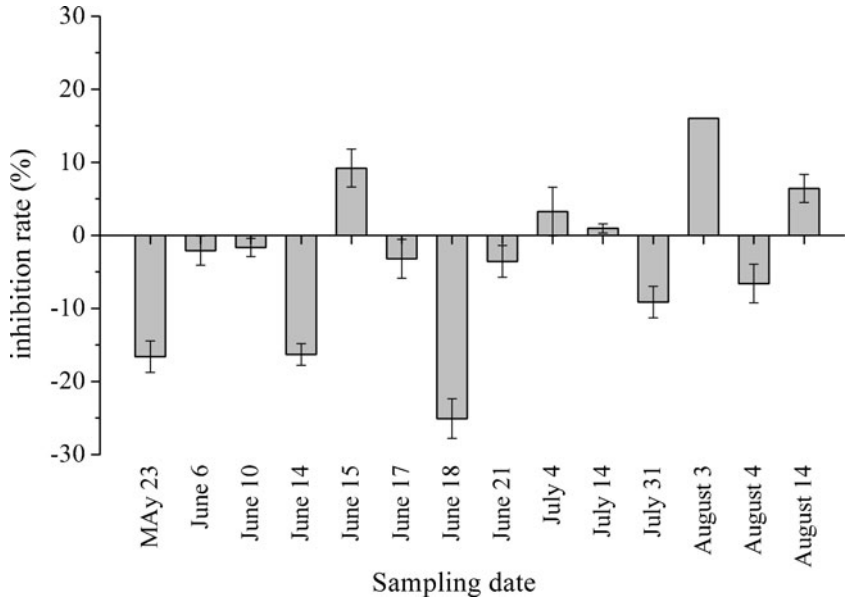

Fig. 14 Inhibition rate of *Q67* exposed to composite samples of urban stormwater runoff (May 23–August 4, 2011 storm events)

Table 2 Pearson correlations between concentration of heavy metals and PAHs and toxicology indexes

	Pearson R (*p* value)					
	24 h lethality	Spontaneous movement (24 hpf)	(48 hpf)	Hatching rate (72 hpf)	Abnormality rate (96 hpf)	Inhibition rate (Q67)
As	−0.320	0.014	0.232	0.354	−0.164	−0.502
Se	−0.246	−0.288	−0.242	−0.328	0.665**	−0.058
Zn	0.518*	0.129	−0.253	−0.641**	0.805**	0.830**
Pb	−0.252	0.138	0.200	0.344	−0.152	−0.421
Ni	−0.182	0.339	0.059	0.124	0.508	−0.016
Fe	0.631**	0.140	−0.187	−0.576**	0.236	0.817**
Mn	0.279	0.127	−0.291	−0.546**	0.695**	0.596*
Cr	−0.203	0.366	0.340	0.264	−0.016	−0.183
Cu	0.082	0.017	−0.163	−0.583**	0.881**	0.404
Al	−0.176	−0.045	0.159	0.350	−0.300	−0.139
PAHs	−0.177	−0.418	−0.372	−0.790**	0.721*	0.628

Single asterisk denotes significant correlation at 0.05 level; double asterisks denote significant correlation at 0.01 level

hatching rate and abnormality percentage and Mn concentration were −0.546 ($p<0.01$) and 0.981 ($p<0.01$), respectively. Mn, similar to Zn, plays a number of essential roles in many metabolic functions such as cellular protection and replication mechanisms and bone mineralization processes (Santamaria 2008). Likewise, high Mn level exposure of cell or organisms causes toxicity (CICAD 2004; Pinsino et al. 2010).

Four low-molecular weight (LMW) PAHs were also detected as primary pollutants: naphthalene, fluorine, phenanthrene, and pyrene. Despite of the low concentration of PAHs in the stormwater runoff samples, the concentration was significantly correlated with hatching and abnormality rate, as shown in Table 2, though PAHs were surely not solely responsible for the total toxicity. Previous studies indicate the 2∼4 ring PAHs can cause the zebrafish embryos teratogenesis (including pericardial edema, yolk sac edema, lordosis, and scoliosis), cardiac dysfunction, neuronal death, and so on (Carls et al. 1999). Due to the lipophilic characteristics, PAHs may easily enter the larval body during the development of embryo, especially after losing the chorionic barrier function, resulting in abnormal chromosomal development, disturbed metabolic processes, abnormal tissue differentiation, low physiological functions, ultimately not hatching and teratogenesis (Carls et al. 1999).

The inhibition rate of *Q67* luminous affected by urban stormwater runoff was from 0.96 (±0.63) % to 60.99 (±6.06) % (Fig. 13). Ma et al. assessed the acute toxicity of the river water samples with *Q67* (Ma et al. 1999). The result showed the inhibition rates of samples varied from 6.39 %∼83.56 % and the river water was seriously polluted by heavy metals. However, the toxic effect of runoff samples on the *Q67* was complicated, which was not caused by one single factor, but by multiple pollutants together. Data from statistical analysis showed that, the main factors affected the inhibition rate of *Q67* luminous intensity were zinc, iron, and manganese (Table 2). The process of bioluminescence involves luciferase and several other related enzymes (e.g., electron transport; Thomulka et al. 1993). These enzyme systems are inhibited by either one or both of two possible mechanisms: direct inhibition of luminescence activity or indirect inhibition of luminescence by general cytoplasmic poisons which interfere with essential and related metabolic processes. Reduced flavin mononucleotide ($FMNH_2$) plays a key role in the luminescence reaction. During the luminescence reaction, there are

Table 3 Pearson correlations between biotoxicology indexes

Biotoxicology indexes	Pearson R (*P* value)				
	Spontaneous movements (24 hpf)	(48 hpf)	Hatching rate (72 hpf)	Percent abnormality (96 hpf)	Inhibition rate (Q67)
24 h lethality	0.013	−0.366	−0.687**	−0.355	0.848**
Spontaneous movements (24 hpf)	1	0.519*	0.513*	−0.186	−0.094
Heart beat rate (48 hpf)		1	0.542*	−0.163	−0.462
Hatching rate (72 hpf)			1	−0.062	−0.806**
Percent abnormality (96 hpf)				1	−0.017

Single asterisk denotes significant correlation at 0.05 level; double asterisks denote significant correlation at 0.01 level

a series of chemical changes between heavy metals and riboflavin or ribose side chain, which produce anti-riboflavin compounds and inhibit the activity of flavin mononucleotide. In addition, compared with the control group, about 64 % of composite runoff samples produced a stimulatory effect on the *Q67*, resulting in higher luminescence (Fig. 14). The phenomenon is called hormesis-effect which characterized by a low-dose stimulation and a high-dose inhibition. However, the mechanisms underlying hormesis induced by environmental agents are not well elucidated. The current explanation about hormesis was that it is due to an overcompensation in response to a disruption in homeostasis and this was supported through experimentation (Calabrese and Howe 1976). The compensatory response may eventually exceed the performance of the controls, resulting in the net stimulatory response commonly referred to as a hormetic effect (Calabrese 1999). This phenomenon has been observed in a number of studies (Kayhanian et al. 2008), which indicated that hormesis is another effect of urban stormwater runoff on *Q67*.

In this study, the data of grab samples showed that the toxic effects of runoff on hatching and abnormality of zebrafish embryos and inhibition of *Q67* luminescence were stronger at the early 15 min than those at later stage. This might be explained with the first flush in urban areas. The storm events with grab samples had short and heavy flow, and the concentration peak was always followed by the flow peak (Lee and Bang 2000) indicating that adequate pollutants could inhibit or damage the growth of zebrafish embryos and *Q67*.

Besides, the magnitude of toxicity responses for zebrafish embryos of 24 h lethality and 72 h hatching rate and inhibition rate of *Q67* were significantly correlated with each other (Table 3). There were also significant correlations between spontaneous movements, and 72 h hatching rate. However, 96 h abnormality was not significantly correlated with the others (Table 3), indicating that different toxicants could have varied effects on the same organism (Schiff et al. 2002).

Conclusion

Urban highway runoff from seventeen storm events had negative effects on the development of zebrafish embryos and *Q67*. Lethal, sublethal, and teratogenic endpoints and luminous intensity of *Q67* were recorded in the zebrafish embryos tests and *Q67* acute toxicity test. Toxicity to zebrafish embryos and *Q67* from runoff samples was frequently observed. Stronger toxicity was observed of the runoff collected at the initial rainfall stage. Compared with the control group, urban highway runoff significantly increased the lethality, reduced the percent with spontaneous movements (24 hpf) in 20 s and heart beats (48 hpf), inhibited the embryos hatching (72 hpf) and induced development abnormalities (96 hpf) of zebrafish embryo, and inhibited or promoted the luminescence of *Q67*. Among all the pollutants, heavy metals (especially copper, manganese, and zinc) were the primary cause of toxicity. Overall, urban highway runoff in Shanghai has the potential ecotoxicity to aquatic organisms. Therefore, more attention should be paid to remove the toxic fraction of pollutants, especially to control the concentrations of heavy metals in the urban highway runoff.

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